ILAB: OeNB

Documentation

Variational Autoencoders

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Inhaltsverzeichnis

1 Agenda **2**
1.1 Agenda 2

2 Out-of-Sample **3**
2.1 Ratios Only 3
2.1.1 Training Set 3
2.2 Time Series 3
2.2.1 Training Set 3
2.2.2 Test Set 4
2.3 Sector Deviations 4
2.3.1 Training Set 4
2.3.2 Test Set 4
2.4 All 5
2.4.1 Training Set 5
2.4.2 Test Set 5

Kapitel 1

Agenda

1.1 Agenda

- **Research Question 1: Develop and evaluate supervised credit scoring models for Austrian SMEs using raw balance-sheet data.**
  - **VAE Integration:** Does the integration of VAEs improve the CV-AUC?
  - **AutoML:** Can AutoML lead to better results?
  - **Model Selection:** Leaderboard Results and Strategy Comparison (GLM, RF and Boosting).

- **Research Question 2: Does the use of financial ratios improve model performance and generalization compared to the raw-data approach?**
  - **Model Training:** Model Comparison and identifing the optimal hyperparameters.
  - **Model Selection:** Which model is finally selected?
  - **Comparative Analysis:** Raw accounting-positions versus financial ratios.
- **Outlook: Research Question 3 and Year specifications.**

Kapitel 2

Out-of-Sample

2.1 Ratios Only

2.1.1 Training Set

Tabelle 2.1: XGBoost (5-Fold CV) - Base Model

Rank	η	**Depth**	**Subsample**	**Colsample**	**MCW**	γ	**L2** (λ)	**L1** (α)	**CV AUC**
1	0.0715	4	0.773	0.933	4	2.700	2.960	1.740	**0.827**
2	0.2980	3	0.690	0.566	20	0.191	0.000	4.900	0.823
3	0.0934	5	0.970	0.764	11	4.780	3.390	0.515	0.822

Note: The optimal configuration converged at 83 boosting rounds.

In-sample AUC: 0.8717 | *Total tuning time:* 92.2s.

Tabelle 2.2: XGBoost (5-Fold CV) - Dimensionality Reduction

Rank	η	**Depth**	**Subsample**	**Colsample**	**MCW**	γ	**L2** (λ)	**L1** (α)	**CV AUC**
1	0.0323	5	0.665	0.958	3	0.238	3.513	4.327	**0.755**
2	0.1540	8	0.989	0.872	8	5.000	0.028	5.000	0.752
3	0.0731	4	0.796	0.754	16	0.862	2.770	3.210	0.750

Note: The optimal configuration converged at 183 boosting rounds.

In-sample AUC: 0.8326 | *Total tuning time:* 142.4s.

2.2 Time Series

2.2.1 Training Set

Tabelle 2.3: XGBoost Top 3 Optimised Configurations (5-Fold CV) - Time Series Features

Rank	η	**Depth**	**Subsample**	**Colsample**	**MCW**	γ	**L2** (λ)	**L1** (α)	**CV AUC**
1	0.0715	4	0.773	0.933	4	2.700	2.960	1.740	**0.900**
2	0.0934	5	0.970	0.764	11	4.780	3.390	0.515	0.899
3	0.3000	3	0.784	0.961	18	1.520	0.000	3.650	0.897

Note: The optimal configuration converged at 97 boosting rounds.

In-sample AUC: 0.9434 | *Total tuning time:* 339.1s.

Tabelle 2.4: XGBoost Test Set Performance (Time Series Model)

Evaluation Metric	**Value**
Probabilistic Metrics	
Area Under the ROC Curve (AUC)	0.9083
Brier Score	0.1052
Penalised Brier Score	0.1712
Log Loss	0.3315
Classification Metrics (Threshold = 0.5560)	
Sensitivity (Recall)	0.7958
Specificity	0.8690
Precision	0.0484
F1-Score	0.0912

Note: Hard classification metrics were evaluated using the Youden-optimal probability threshold of 0.5560.

2.2.2 Test Set

2.3 Sector Deviations

2.3.1 Training Set

Tabelle 2.5: XGBoost Top 3 Optimised Configurations (5-Fold CV)

Rank	η	**Depth**	**Subsample**	**Colsample**	**MCW**	γ	**L2** (λ)	**L1** (α)	**CV AUC**
1	0.0268	6	0.650	0.796	8	1.140	4.590	4.750	**0.823**
2	0.0934	5	0.970	0.764	11	4.780	3.390	0.515	0.821
3	0.0532	7	0.867	0.628	18	1.700	1.070	1.430	0.820

Note: The optimal configuration converged at 136 boosting rounds.

In-sample AUC: 0.9158 | *Total tuning time:* 196.3s.

2.3.2 Test Set

Tabelle 2.6: XGBoost Test Set Performance

Evaluation Metric	**Value**
Probabilistic Metrics	
Area Under the ROC Curve (AUC)	0.8270
Brier Score	0.1308
Penalised Brier Score	0.2403
Log Loss	0.4033
Classification Metrics (Threshold = 0.4699)	
Sensitivity (Recall)	0.7197
Specificity	0.7810
Precision	0.0269
F1-Score	0.0518

Note: Hard classification metrics were evaluated using the Youden-optimal probability threshold of 0.4699.

2.4 All

2.4.1 Training Set

Tabelle 2.7: XGBoost Top 3 Optimised Configurations (5-Fold CV)

Rank	η	**Depth**	**Subsample**	**Colsample**	**MCW**	γ	**L2** (λ)	**L1** (α)	**CV AUC**
1	0.0934	5	0.970	0.764	11	4.780	3.390	0.515	**0.907**
2	0.1310	5	0.912	0.742	17	2.750	0.115	1.760	0.905
3	0.2960	4	0.663	1.000	16	0.137	2.290	4.000	0.898

Note: The optimal configuration converged at 69 boosting rounds.

In-sample AUC: 0.9656 | *Total tuning time:* 242.7s.

2.4.2 Test Set

Tabelle 2.8: XGBoost Test Set Performance (Final Revised Model)

Evaluation Metric	**Value**
Probabilistic Metrics	
Area Under the ROC Curve (AUC)	0.9160
Brier Score	0.0849
Penalised Brier Score	0.1448
Log Loss	0.2784
Classification Metrics (Threshold = 0.4967)	
Sensitivity (Recall)	0.8114
Specificity	0.8810
Precision	0.0539
F1-Score	0.1011

Note: Hard classification metrics were evaluated using the Youden-optimal probability threshold of 0.4967.